Genus

Bacteroidetes/Bacteroidia/Bacteroidales/Porphyromonadaceae/

Porphyromonas

Shah and Collins 1988, 129[VP] emend. Willems and Collins 1995, 580.

Paula Summanen, *VA Medical Center West Los Angeles, Anaerobe Laboratory, 11301 Wilshire Boulevard, Building 304, Room E3-237, Los Angeles, CA 90073, USA*

Sydney M. Finegold, *VA Medical Center West Los Angeles, Infectious Diseases Section (111 F), Los Angeles, CA 90073, USA*

Por.phy.ro.mo'nas. Gr. adj. *porphyreos* purple; Gr. n. *monas* unit; N.L. fem. n. *Porphyromonas* porphyrin cell.

Short rods or coccobacilli, 0.3–1 × 0.8–3.5 μm. Gram-negative, non-sporeforming, and nonmotile. **Obligately anaerobic**. Generally cells form **brown to black colonies on blood agar due to protoheme production**. Most species are asaccharolytic: growth is not significantly affected by carbohydrates but is enhanced by protein hydrolysates such as proteose peptone or yeast extract. **Major fermentation products are usually n-butyric acid and acetic acid**; propionic, isovaleric, isobutyric, and phenylacetic acid may also be produced. The major cellular fatty acid is 13-methyltetradecanoic acid ($C_{15:0}$ iso). Indole is produced by most strains. Nitrate is not reduced to nitrite. Esculin is not hydrolyzed. Most species do not hydrolyze starch. Isolated from oral infections and various other clinical specimens of human and animal origin.

DNA G+C content (mol%): 40–55 (T_m).

Type species: **Porphyromonas asaccharolytica** (Holdeman and Moore 1970) Shah and Collins 1988, 128[VP] [*Bacteroides asaccharolyticus* (Holdeman and Moore 1970) Finegold and Barnes 1977, 390; *Bacteroides melaninogenicus* subsp. *asaccharolyticus* Holdeman and Moore 1970, 33].

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Number of validated species: 16

Further descriptive information

Cell morphology

Most cells in broth cultures are small (0.3–1.0 × 0.8–3.5 μm); however, occasionally longer cells and filaments (≥5 μm) may

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be formed. Cells from growth on a solid medium are commonly shorter and can appear spherical.

Cell-wall composition

The cell-wall peptidoglycan contains lysine as the diamino acid. 2-Keto-3-deoxyoctulosonic acid is absent. The principal respiratory quinones are unsaturated menaquinones with 9 or 10 isoprene units. Both nonhydroxylated and 3-hydroxylated fatty acids are present. The nonhydroxylated fatty acids are predominantly methyl-branched-chain fatty acids. The predominant fatty acid is $C_{15:0}$ iso; a few species contain comparable amounts of $C_{15:0}$ iso and $C_{15:0}$ anteiso acids. The 3-hydroxylated fatty acids are generally straight-chain saturated fatty acids. Information on the fatty acid content and cell wall composition can be found in Collins et al. (1994) and Brondz and Olsen (1991).

Colony morphology

Colonies on blood agar plates are usually round, entire, smooth (occasionally rough), shiny, convex, and 0.5–3 mm in diameter. The colonies of all but one species (*Porphyromonas catoniae*) are pigmented. The black pigmentation of *Porphyromonas* is caused by the accumulation of hemin used as an iron source for bacterial growth. The species vary in the degree and rapidity of pigment production depending primarily on the type of blood used in the growth medium. Laked rabbit blood agar is considered the most reliable medium for detecting the pigment. The pigmentation ranges from tan to black and may take several days to develop.

Growth conditions

The optimum temperature for growth is 37°C. *Porphyromonas* species favor a slightly alkaline environmental pH and 100% humidity. Hemin and vitamin K_1 are either required or greatly stimulate the growth of most species. Although some species are weakly saccharolytic, their growth is not significantly affected by carbohydrates. Nitrogenous substances, such as proteose peptone, trypticase, and yeast extract markedly enhance growth.

Metabolism and metabolic pathways

Malate dehydrogenase and glutamate dehydrogenase are present; glucose-6-phosphate dehydrogenase and 6-phosphogluconate dehydrogenase are absent from most species. Proteolytic activity is variable. The strains have a limited ability to ferment amino acids such as aspartate and asparagine.

Genetics

Porphyromonas forms a distinct phylogenetic group, and 16S rRNA gene sequencing can be reliably used to differentiate *Porphyromonas* from other genera and from each other, with the exception of *Porphyromonas asaccharolytica* and *Porphyromonas uenonis*, where DNA–DNA reassociation studies are required to distinguish these two species genetically.

The genome of *Porphyromonas gingivalis* has been sequenced and studied by the Forsyth Institute and The Institute for Genomic Research (TIGR) (Nelson et al., 2003). The genome size of this species was determined to be 2176 kb from *Xba*I restriction enzyme digests and 2250 kb from *Spe*I digests. Information and a schematic representation of the *Porphyromonas gingivalis* W83 genome can be found at www.tigr.org. Comparative analysis of the whole-genome sequence with other available complete genome sequences confirmed the close relationship between the phylum *Bacteroidetes* [*Cytophaga–Flavobacteria–Bacteroides* (CFB)] and the green sulfur bacteria. Within the phylum "*Bacteroidetes*" the genomes of *Bacteroides thetaiotaomicron* and *Bacteroides fragilis* were most similar to that of *Porphyromonas gingivalis*. The genome analysis revealed a range of virulence determinants that relate to the novel biology of this bacterium. It also revealed that *Porphyromonas gingivalis* can metabolize a range of amino acids and generate end products that are toxic to the human host.

Antibiotic susceptibility

Porphyromonas species are generally very susceptible to most of the antimicrobial agents commonly used for treatment of anaerobic infections, such as amoxicillin-clavulanate, piperacillin-tazobactam, ampicillin-sulbactam, imipenem, cephalosporins, and metronidazole. β-Lactamase production has been described in *Porphyromonas asaccharolytica* (Aldridge et al., 2001), *Porphyromonas catoniae* (Kononen et al., 1996), *Porphyromonas somerae* (Summanen et al., 2005), and *Porphyromonas uenonis* (Finegold et al., 2004). The frequency of β-lactamase production has been reported at approximately 20%. Animal-derived *Porphyromonas* species are more often β-lactamase producers than those derived from humans. Occasional resistance to clindamycin and ciprofloxacin may occur in *Porphyromonas asaccharolytica* and *Porphyromonas somerae*; also, ciprofloxacin resistance of *Porphyromonas gingivalis* has been reported (Lakhssassi et al., 2005).

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Pathogenicity

Some species are considered true pathogens and are associated with human or animal infections. In particular, *Porphyromonas gingivalis* is a major causative agent in the initiation and progression of severe forms of periodontal disease. *Porphyromonas gingivalis* possesses a multitude of cell-surface associated and extracellular activities that contribute to its virulence potential. Several of these factors are adhesins that interact with other bacteria, epithelial cells, and extracellular matrix proteins. Secreted or cell-bound enzymes, toxins, and hemolysins play a significant role in the spread of the organism through tissue, in tissue destruction, and in evasion of host defenses.

Ecology

Several of the members of *Porphyromonas* are indigenous bacterial flora in the oral cavity of humans and animals. Many species are also found in the urogenital and intestinal tracts. *Porphyromonas* species have been isolated from oral infections, and from many infections throughout the body, e.g., blood, amniotic fluid, umbilical cord, pleural empyema, peritoneal and pelvic abscesses, inflamed endometrium, and other infected tissues. *Porphyromonas* species of animal origin have been encountered in humans with animal bite infections.

Enrichment and isolation procedures

A complex medium containing peptone, yeast extract, vitamin K_1, and hemin, and supplemented with 5% blood is recommended for isolation of *Porphyromonas* from body sites. Fresh or prereduced media (commercially available from Anaerobe Systems, Morgan Hill, California) are recommended, as they increase isolation efficacy. For determination of pigmentation, rabbit blood (laked) is preferable to blood from other animals. Kanamycin-vancomycin laked-blood agar with reduced vancomycin concentration (2 μg/ml) may be used as a selective medium when *Porphyromonas* species are sought (Jousimies-Somer et al. 2002). Formulas for basal media are given in Jousimies-Somer et al. (2002).

Inoculated media should immediately be placed in an anaerobic environment, such as an anaerobic pouch, jar, or chamber. If anaerobic systems utilizing palladium catalysts are used, the anaerobic gas mixture must contain hydrogen to enable the palladium to reduce oxygen to water. The plates may be examined after a 48-h incubation. However, a total incubation period of at least 7 d is recommended, because not all *Porphyromonas* species may be detected with shorter incubation times.

Maintenance procedures

Isolates can be put into stock from broth or plate cultures. Freezing at −70°C or lyophilization of young cultures grown

TABLE 1. Differentiation of the genus *Porphyromonas* from other anaerobic Gram-negative rods[a]

Characteristic	*Porphyromonas*[b]	*Alistipes*[c]	*Bacteroides*[d]	*Prevotella*	*Rikenella*	*Tannerella*
Susceptibility to:[e]						
Vancomycin (5 μg)	S	R	R	R	R	R
Kanamycin (1000 μg)	R	R	R	R	S	S
Colistin (10 μg)	R	R	R	D	R	S
Pigment	+	+	−	D	−	−
Growth in 20% bile	−	+	+	−	+	−
Proteolytic activity	D	+	−	D	−	+
Major metabolic end products from PYG[f]	A, B, iV	S	A, S	A, S	P, S	A, S, PA
Major long-chain fatty acids	$C_{15:0}$ iso	$C_{15:0}$ iso	$C_{15:0}$ anteiso	$C_{15:0}$ anteiso	$C_{15:0}$ iso	$C_{15:0}$ anteiso
DNA G+C (mol%)	40–55	55–58	40–48	39–60	60–61	44–48

[a]+, 90% or more of the strains are positive; −, 10% or more of the strains are negative; D, different reaction in different species.
[b]*Porphyromonas catoniae* does not produce pigment, is vancomycin-resistant, and fermentative.
[c]*Alistipes putredinis* does not produce pigment, is susceptible to bile, and nonfermentative.
[d]*Bacteroides sensu stricto.*
[e]Special potency antimicrobial identification disks; R, resistant; S, susceptible; D, differs among species.
[f]A, acetic acid; B, butyric acid; iV, isovaleric acid; P, propionic acid; PA, phenylacetic acid; S, succinic acid.

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in a well-buffered liquid or solid medium is satisfactory for storage of *Porphyromonas* species. Storage of lyophilized cultures at 4°C is recommended. Even with the best storage conditions, only a portion of the original cell population survives. Therefore, a large inoculum in a supportive medium and minimal exposure to oxygen are recommended for the recovery of viable cultures from stored material.

To maintain stock strains in the laboratory, it is advisable to transfer them weekly in chopped meat medium or other suitable medium that does not contain a fermentable carbohydrate.

Differentiation of the genus *Porphyromonas* from other genera

The pigmented *Porphyromonas* species can be differentiated from other anaerobic, Gram-stain-negative genera with relative ease (Table 1). The special-potency antibiotic disks (vancomycin, 5 μg; kanamycin, 1000 μg; and colistin, 10 μg) can be used to separate the Gram-stain-negative genera: *Porphyromonas* species are generally sensitive to vancomycin and resistant to kanamycin and colistin, whereas the other Gram-stain-negative genera are resistant to vancomycin and vary in their resistance to kanamycin and colistin.

A distinctive feature of the porphyromonads is that the predominant fatty acid is 13-methyltetradecanoic acid ($C_{15:0}$ iso); by contrast, *Bacteroides* and *Prevotella* contain 12-methyl tetradecanoic acid ($C_{15:0}$ anteiso) as their major cellular fatty acid. *Bacteroides* and *Prevotella* also produce a simpler metabolic end product profile of mainly acetic and succinic acids (Table 1).

Taxonomic comments

The taxonomy of pigmented Gram-negative bacilli has changed greatly since the first edition of *Bergey's Manual of Systematic Bacteriology*. The genus *Porphyromonas* was created in 1988 (Shah and Collins, 1988), and since then some species previously included in the genus *Bacteroides* (*Porphyromonas gingivalis*, *Porphyromonas asaccharolytica*, *Porphyromonas endodontalis*, *Porphyromonas levii*, and *Porphyromonas macacae*) were reclassified as *Porphyromonas* species, and several new species described. *Porphyromonas* now includes 16 validly published species. *Porphyromonas salivosa* is a later heterotypic synonym of *Porphyromonas macacae* (Love, 1995).

The porphyromonads form a natural, but deep, phylogenetic group (Figure 1). The species exhibit levels of 16S rRNA sequence divergence of up to 15%. Phylogenetically, the closest related species to *Porphyromonas* are found in the genera *Bacteroides* and *Prevotella* (approx. 82–89% similarity).

Differentiation of species of the genus *Porphyromonas*

With the exception of *Porphyromonas catoniae*, *Porphyromonas* species form a phenotypically homogeneous group: they are pigmented, sensitive to the special-potency vancomycin disk, and produce butyric acid as the major metabolic end product. *Porphyromonas catoniae* differs in all these aspects. The characteristics useful in differentiating the species are given in Table 2.

Numerous PCR-based identification or detection systems have been described for *Porphyromonas gingivalis* and *Porphyromonas endodontalis* (de Lillo et al., 2004; Fouad et al., 2002; Gomes et al., 2005; Jervoe-Storm et al., 2005; Kuboniwa et al., 2004; Kumar et al., 2003; Noguchi et al., 2005; Seol et al., 2006).

List of species of the genus *Porphyromonas*

Porphyromonas asaccharolytica (Holdeman and Moore 1970) Shah and Collins 1988, 129[VP] (*Bacteroides asaccharolyticus* (Holdeman and Moore 1970) Finegold and Barnes 1977, 390[AL]; *Bacteroides melaninogenicus* subsp. *asaccharolyticus* Holdeman and Moore 1970, 33)

a.sac.cha.ro.ly'ti.cus. Gr. pref. *a* not; Gr. n. *sakchâr* sugar; N.L. fem. adj. *lytica* (from Gr. fem. adj. *lutikê*) able to loosen, able to dissolve; N.L. fem. adj. *asaccharolytica* not digesting sugar.

The description is from Shah and Collins (1988) and Holdeman et al. (1984). Proteolytic activity is weak, but gelatin liquefaction is positive and fibrinolytic activity is present. Starch is not hydrolyzed. Growth is stimulated by 0.5% NaCl. Cellular and colonial morphology and other characteristics are as described for the genus and as given in Table 2. *Porphyromonas asaccharolytica* can be distinguished from *Porphyromonas uenonis* by phenotypic tests (Table 2); however, DNA–DNA reassociation studies are required to distinguish these two species genetically. Susceptible to piperacillin-tazobactam, ampicillin-sulbactam, imipenem, meropenem, trovafloxacin, and metronidazole. Most strains (≥90%) are susceptible to cefoxitin, ciprofloxacin, and clindamycin. β-Lactamase production has been described in *Porphyromonas asaccharolytica* at the rate of 21% (Aldridge et al., 2001).

Source: various human clinical infections.

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FIGURE 1. Unrooted tree showing the phylogenetic position of *Porphyromonas* within the *Bacteroides* subgroup of the *Bacteroidetes* (*Cytophaga–Flavobacter–Bacteroides*) phylum. The tree was constructed by the maximum-parsimony method and is based on a comparison of approximately 1400 nt. Bootstrap values, expressed as a percentage of 1000 replications, are given at the branching points. The scale bar indicates 1% sequence divergence. Courtesy of Paul A. Lawson.

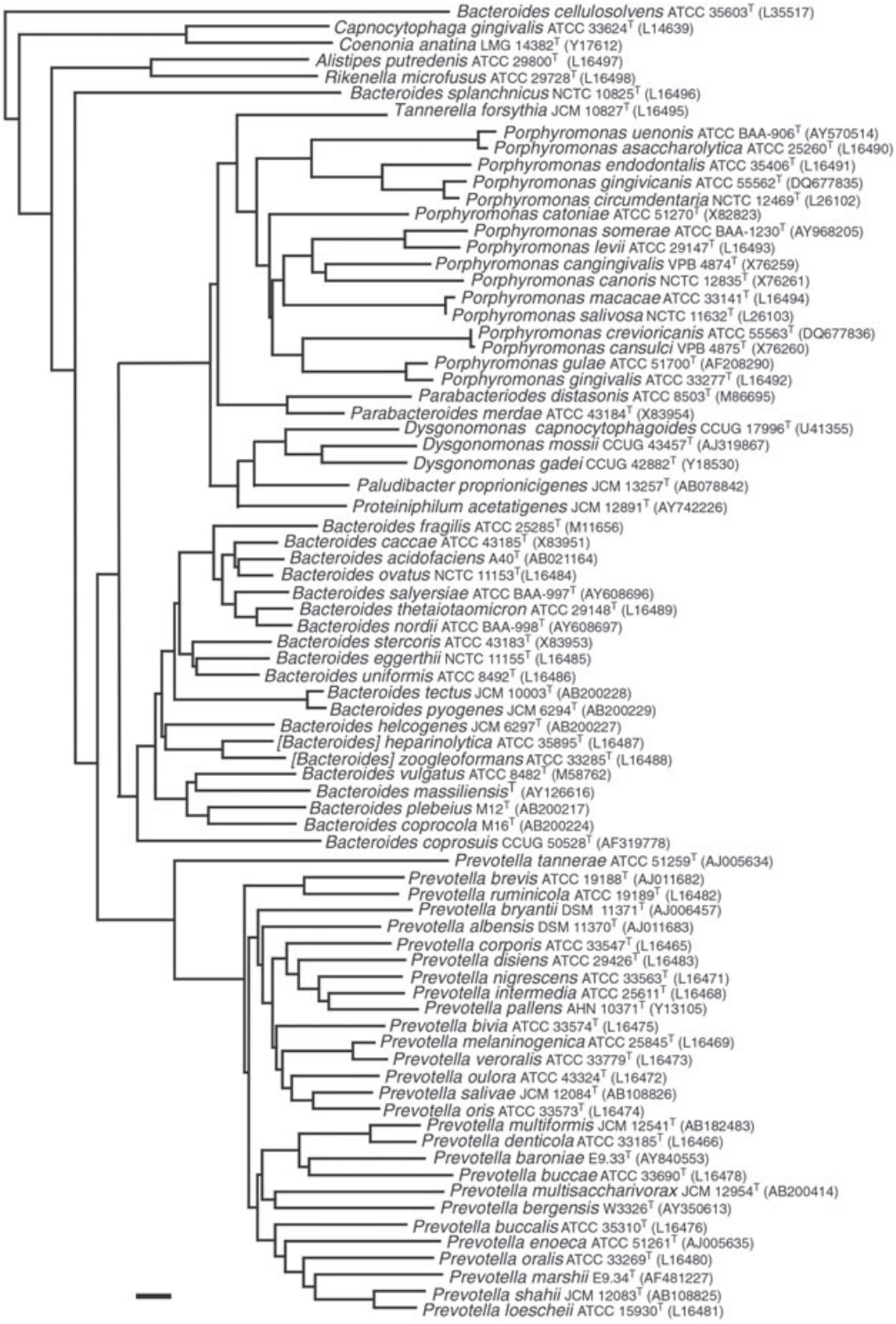

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TABLE 2. Differentiation of the species of the genus *Porphyromonas*[a]

Characteristic	*P. asaccharolytica*	*P. canginginvalis*	*P. canoris*	*P. cansulci*	*P. catoniae*	*P. circumdentaria*	*P. crevioricanis*	*P. endodontalis*	*P. gingivalis*	*P. gingivicanis*	*P. gulae*	*P. levii*	*P. macacae*[e]	*P. somerae*	*P. uenonis*
Pigment															
production	+	+	+	+	−	+	+	+	+	+	+	+	+	+	+
Fluorescence	+	−	+	+	−	+	+	+	−	+	−	d	−	d	+
Hemagglutinin															
activity	−	−	−	−	na	−	+	−	+	−	+	−	−	na	na
Indole	+	+	+	+	−	+	+	+	+	+	+	−	+	−	+
Catalase	−	+	+	+	−	+	−	−	−	+	+	−	+	−	−
Lipase	−	−	−	−	−	−	−	−	−	−	−	−	+[−]	−	−
Preformed enzyme activity:[b]															
α-Fucosidase	+	−	−	−	+	−	−	−	−	−	−	−	−	−	−
α-Galactosidase	−	−	−	−	d	−	−	−	−	−	−	−	+	−	−
β-Galactosidase	−	−	+	−	+	−	−	−	−[f]	−	+	+	−[f]	+	−
N-Acetyl-β-glucosaminidase	−	−	+	−	+	−	−	−	+	−	+	+	+	+	d
Chymotrypsin	−	+	+	−	d	+	−	−	−	−	na	+	+	+	−
Trypsin	−	−	−	−	d	−	−	−	+	−	+	−	+	−	−
Fermentation of:[c]															
Glucose	−	−	−	−	+	−	−	−	−	−	−	w	w	w	−
Lactose	−	−	−	−	+	−	−	−	−	−	−	w	w	w	−
Maltose	−	−	−	−	+	−	−	−	−	−	−	w	−	w	−
Glucose-6-phosphate and 6-phosphogluconate dehydrogenases present	−	+	+	−	+	−	−	−	−	−	na	d	d	na	na
Major long-chain fatty acids	$C_{15:0}$ iso	$C_{15:0}$ iso	$C_{15:0}$ iso	$C_{15:0}$ iso	$C_{15:0}$ iso, $C_{15:0}$ anteiso	$C_{15:0}$ iso	$C_{15:0}$ iso	$C_{15:0}$ iso	$C_{15:0}$ iso	$C_{15:0}$ iso	na	$C_{15:0}$ iso, $C_{15:0}$ anteiso	$C_{15:0}$ iso	$C_{15:0}$ iso, $C_{15:0}$ anteiso	$C_{15:0}$ iso
Metabolic end products from PYG[d]	A, P, ib, B, IV, s	A, p, ib, B, IV	A, P, ib, b, IV, s	A, P, ib, B, IV, S, pa	a, P.iv, l, S	A, P, ib, b, IV, s, pa	A, p, ib, B, IV, s, pa	A, P, ib, B, IV, s	A, P, ib, B, IV, s, pa	A, p, ib, B, IV, s	A, P, ib, B, IV, s, pa	A, P, ib, B, IV, s	A, P, ib, B, IV, s, pa	A, P, ib, B, IV, s	A, P, ib, B, IV, s

[a]+, 90% or more of the strains are positive; −, 10% or more of the strains are negative; d, 11–89% of the strains are positive; w, weak positive reaction; na, data not available.

[b]Reaction by API ZYM System or Rosco Diagnostic tablets. Reactivity in these systems is not always identical (see footnote f).

[c]Fermentation of most other carbohydrates have been reported negative.

[d]Upper-case letters indicate major metabolic products from peptone-yeast-glucose (PYG), lower-case letters indicate minor products, and parentheses indicate a variable reaction for the following acids: A, acetic; P, propionic; IB, isobutyric; B, butyric; IV, isovaleric; V, valeric; L, lactic; S, succinic; PA, phenylacetic.

[e]*Porphyromonas salivosa* is a later heterotypic synonym. The cat biovar (*Porphyromonas salivosa*) is lipase-positive, does not ferment sorbitol, and may fluoresce under UV light.

[f]Negative by the API ZYM System; positive with the Rosco *o*-nitrophenyl-β-D-galactopyranoside test.

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DNA G+C content (*mol%*): 52–54 (T_m)

Type strain: ATCC 25260, CCUG 7834, DSM 20707, JCM 6326, LMG 13178, L16490, VPI 4198.

Sequence accession no. (*16S rRNA gene*): L16490.

Porphyromonas cangingivalis Collins, Love, Karjalainen, Kanervo, Forsblom, Willems, Stubbs, Sarkiala, Bailey, Wigney and Jousimies-Somer 1994, 676[VP]

can.gin.gi.val'is. L. n. *canis* dog; L. n. *gingiva* gum; L. fem. suff. *-alis* suffix denoting pertaining to; N.L. fem. adj. *cangingivalis* pertaining to the gums of dogs.

The description is from Collins et al. (1994). In cooked-meat carbohydrate medium and on sheep blood agar plates cells are 0.3–0.6 × 0.8–1.5 μm and occur singly and in clumps; occasionally filaments up to 16 μm long are observed. Some strains indent agar, exhibit peripheral flattening, and have a roughened and dry surface appearance on sheep blood agar. On egg yolk agar, colonies are yellow or orange. After incubation of 5 d, the pH range in media containing carbohydrates generally is 6.3–6.5. Ammonia is produced in cooked meat medium. Neither lactate nor threonine is converted to propionate, and pyruvate is not utilized. Other characteristics are as described for the genus and as given in Table 2. Some strains are susceptible to penicillin, amoxicillin, carbenicillin, and erythromycin; 35% of the strains produce β-lactamase.

Source: diseased or healthy periodontal pockets of dogs with naturally occurring periodontitis.

DNA G+C content (*mol%*): 55 (T_m).

Type strain: ATCC 700135, CCUG 47700, NCTC 12856, VPB 4874, X76259.

Sequence accession no. (*16S rRNA gene*): X76259.

Porphyromonas canoris Love, Karjalainen, Kanervo, Forsblom, Willems, Stubbs, Sarkiala, Bailey, Wigney and Jousimies-Somer 1994, 207[VP]

can'or.is. L. n. *canis* dog; L. gen. n. *oris* of the mouth; N.L. gen. n. *canoris* of a dog's mouth.

The description is from Love et al. (1994). In cooked-meat carbohydrate medium and on sheep blood agar plates cells are 0.3–0.6 × 0.8–1.5 μm and occur singly and in clumps; occasionally filaments up to 16 μm long are observed. On sheep blood agar, colonies at 48 h are circular and rough, and have an orange pigmentation. After incubation of 5 d, the pH range in media containing carbohydrates is 6.3–6.5. Ammonia is produced in cooked meat medium. Lactate is converted to propionate, but pyruvate is not utilized and threonine is not converted to propionate. Other characteristics are as described for the genus and as given in Table 2. Strains are susceptible to penicillin, amoxicillin, carbenicillin, and erythromycin.

Source: subgingival pockets of dogs with naturally occurring periodontitis.

DNA G+C content (*mol%*): 49–51 (T_m).

Type strain: CCUG 36550, CIP 104881, JCM 11138, NCTC 12835, VPB 4878.

Sequence accession no. (*16S rRNA gene*): X76261.

Porphyromonas cansulci Collins, Love, Karjalainen, Kanervo, Forsblom, Willems, Stubbs, Sarkiala, Bailey, Wigney and Jousimies-Somer 1994, 678[VP]

can.sul'ci. L. n. *canis* dog; L. gen. n. *sulci* of a furrow; L. gen. n. *cansulci* of a dog's furrow, referring to the habitat in the mouths of dogs.

The description is from Collins et al. (1994). On egg yolk agar, colonies may be yellow or orange. After incubation of 5 d, the pH range in media containing carbohydrates is 6.3–6.5. Ammonia is produced in cooked-meat medium. Neither lactate nor threonine is converted to propionate and pyruvate is not utilized. Other characteristics are as described for the genus and as given in Table 2. Strains are susceptible to penicillin, amoxicillin, carbenicillin, and erythromycin. β-Lactamase producing strains have not been detected.

Source: periodontal pockets of dogs with naturally occurring periodontitis.

DNA G+C content (*mol%*): 49–51 (T_m).

Type strain: CCUG 47702, NCTC 12858, VPB 4875.

Sequence accession no. (*16S rRNA gene*): X76260.

Taxonomic note: Strain ATCC 55563[T] of *Porphyromonas crevioricanis* Hirasawa and Takada 1994, 640[VP] exhibits 99.9% rRNA gene sequence homology with *Porphyromonas cansulci* 12858[T]. The taxonomic standing of these two species remains to be determined.

Porphyromonas catoniae (Moore and Moore 1994) Willems and Collins 1995, 581[VP] (*Oribaculum catoniae* Moore and Moore 1994, 189[VP])

ca.to'ni.ae. N.L. gen. fem. n. *catoniae* of Cato, named in honor of Elizabeth P. Cato, an American microbiologist.

The description is from Willems and Collins (1995) and Moore and Moore (1994). Cells grown in PYG broth

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are 0.6×0.8–1.7 μm and occur in pairs and short chains. Cells grown in media containing fermentable carbohydrate may be highly vacuolated. Surface colonies on blood agar plates incubated for 2 d are 0.2–2 mm in diameter, circular, entire, flat to low convex, and transparent. Most strains are not hemolytic on rabbit blood agar; an occasional strain may be beta-hemolytic. No colonies with a dark pigment are produced. Saccharolytic. Abundant growth occurs in peptone-yeast extract or PYG broth. Broth cultures are turbid with a smooth to fine granular sediment. The pH range of PYG broth cultures is 5.0–5.5. Gelatin is hydrolyzed. Starch is hydrolyzed by most strains; some strains hydrolyze milk and meat. H_2S is produced by some strains. The major cellular fatty acids are $C_{15:0}$ iso and $C_{15:0}$ anteiso; substantial amounts of $C_{13:0}$ iso are also present. Sensitive to clindamycin and chloramphenicol. Two of 85 strains were resistant to erythromycin, one strain to tetracycline, and six strains to penicillin. β-Lactamase producing strains have been described (Kononen et al., 1996).

Source: gingival crevices of humans with gingivitis or periodontitis and from humans with healthy gingivae.

DNA G+C content (*mol%*): 49 (T_m).

Type strain: ATCC 51270, CCUG 41358, NCTC 13056, VPI N3B-3.

Sequence accession no. (*16S rRNA gene*): X82823.

Porphyromonas circumdentaria Love, Bailey, Collings and Briscoe 1992, 435[VP]

cir.cum.den.ta'ri.a. L. adv. and prep. *circum* around, about; L. fem. adj. *dentaria* pertaining to teeth; N.L. fem. adj. *circumdentaria* referring to the isolation of the organism from the vicinity of the teeth.

The description is from Love et al. (1992). In cooked-meat carbohydrate medium and on sheep blood agar plates cells are 0.3–0.6×0.8–1.5 μm and occur singly, in clumps, and occasionally as filaments up to 10 μm long. On sheep blood agar, surface colonies at 72 h are 1–2 mm in diameter, and greenish brown. Both colony size and pigment formation are enhanced by growth in the presence of *Staphylococcus epidermidis*. After incubation for 5 d, the pH range in media containing carbohydrates generally is 6.3–6.5. Ammonia is produced in cooked meat medium. Gelatin is hydrolyzed. Neither lactate nor threonine is converted to propionate, and pyruvate is not utilized. Other characteristics are as described for the genus and as given in Table 2.

Source: soft tissue infections (abscesses and empyemas), gingival margins, and subgingival plaque of felines.

DNA G+C content (*mol%*): 40–42 (T_m).

Type strain: ATCC 51356, CCUG 41934, JCM 13864, NCTC 12469, VPB 3329.

Sequence accession no. (*16S rRNA gene*): L26102.

Porphyromonas crevioricanis Hirasawa and Takada 1994, 640[VP]

cre.vi.o.ri.ca'nis. N.L. gen. n. *crevi* (*sic*) of a crevice; L. gen. n. *oris* of the mouth; L. gen. n. *canis* of a dog; N.L. gen. n. *crevioricanis* of the crevice of a dog's mouth.

The description is from Hirasawa and Takada (1994). Cellular and colonial morphology and other characteristics are as described for the genus and as given in Table 2. Susceptible to penicillin, amoxicillin, sulbenicillin, and erythromycin.

Source: gingival crevicular fluid obtained from beagles.

DNA G+C content (*mol%*): 44–45 (T_m).

Type strain: NUM 402, ATCC 55563.

Sequence accession no. (*16S rRNA gene*): DQ677836.

Taxonomic note: Strain NCTC 12858[T] of *Porphyromonas cansulci* Collins et al. 1994, 678[VP] exhibits 99.9% rRNA gene sequence homology with *Porphyromonas crevioricanis* ATCC 55563[T]. The taxonomic standing of these two species remains to be determined.

Porphyromonas endodontalis (van Steenbergen, VanWinkelhoff, Mayrand, Grenier and de Graaff 1984) Shah and Collins 1988, 129[VP] (*Bacteroides endodontalis* van Steenbergen, VanWinkelhoff, Mayrand, Grenier and de Graaff 1984, 119[VP])

en.do.don'ta.lis. Gr. adv. *endon* within; Gr. n. *odous -ontos* tooth; L. fem. suff. *-alis* suffix denoting pertaining to; N.L. fem. adj. *endodontalis* pertaining to the inside of a tooth, within teeth.

The description is from Shah and Collins (1988) van Steenbergen et al. (1984), and Holdeman et al. (1984). Protoheme is the major porphyrin produced, but protoporphyrin is also present. Most strains produce colonies that adhere strongly to the agar plates, and growth in liquid media is slow.

Gelatin is hydrolyzed. No proteolytic or collagenolytic activity is detected. Arginine is hydrolyzed; starch is not hydrolyzed. H_2S is produced. Cellular and colonial morphology and other characteristics are as described for the genus and as given in Table 2.

Specific primers based on 16S rRNA sequencing have been described for culture-independent identification of *Porphyromonas endodontalis* (de Lillo et al., 2004). Similarly, the use of PCR amplification of 16S rDNA and the downstream

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intergenic spacer region (ISR) for culture-independent detection of periodontal pathogens revealed that *Porphyromonas endodontalis* was as strongly associated with periodontitis as *Porphyromonas gingivalis* (Kumar et al., 2003).

Source: infected dental root canals, periodontal pockets, and other oral sites.

DNA G+C content (*mol%*): 49–51 (T_m).

Type strain: HG370, ATCC 35406, NCTC 13058.

Sequence accession no. (*16S rRNA gene*): L16491.

Porphyromonas gingivalis (Coykendall, Kaczmarek and Slots 1980) Shah and Collins 1988, 129[VP] (*Bacteroides gingivalis* Coykendall, Kaczmarek and Slots 1980, 559[VP])

gin.gi.val'is. L. n. *gingiva* gum; L. fem. suff. -*alis* suffix denoting pertaining to; N.L. fem. adj. *gingivalis* pertaining to the gums, gingival.

The description is from Shah and Collins (1988) and Holdeman et al. (1984). Protoheme is the major porphyrin produced, but traces of protoporphyrin also occur. Several amino acids, such as aspartate, arginine, cystine, histidine, serine, tryptophan, leucine, methionine, phenylalanine, and isoleucine, are utilized. Proteases are present. Starch is not hydrolyzed. Cellular and colonial morphology and other characteristics are as described for the genus and as given in Table 2.

Porphyromonas gingivalis possesses a multitude of cell-surface-associated and extracellular activities, such as adhesins, secreted or cell-bound enzymes, toxins, and hemolysins that contribute to its virulence potential. Differences in the virulence exist, but the mechanisms underlying these differences are not yet fully understood. Multilocus sequence typing of *Porphyromonas gingivalis* strains from different geographic origins showed high genetic diversity of the species (Enersen et al., 2006).

Porphyromonas gingivalis is generally very susceptible to most of the antimicrobial agents commonly used for treatment of anaerobic infections, such as amoxicillin-clavulanate, piperacillin-tazobactam, ampicillin-sulbactam, imipenem, cephalosporins, and metronidazole. Ciprofloxacin resistance of *Porphyromonas gingivalis* has been reported (Lakhssassi et al., 2005).

Porphyromonas gingivalis is considered a major periodontal pathogen and reported to be a cause of extraoral infections, such as lung abscesses and pulmonary infections. It has also been suggested that *Porphyromonas gingivalis* may contribute to the development of atheromas in cardiovascular disease.

Source: infected dental root canals, periodontal pockets, and other oral sites.

DNA G+C content (*mol%*): 46–48 (T_m).

Type strain: 2561, ATCC 33277, CCUG 25893, CCUG 25928, CIP 103683, DSM 20709, JCM 12257, NCTC 11834.

Sequence accession no. (*16S rRNA gene*): L16492.

Porphyromonas gingivicanis Hirasawa and Takada 1994, 639[VP]

gin.gi.vi.ca'nis. L. fem. n. *gingiva* gum; L. gen. n. *canis* of a dog; N.L. gen. n. *gingivicanis* of the gums of a dog.

The description is from Hirasawa and Takada (1994). Cellular and colonial morphology and other characteristics are as described for the genus and as given in Table 2. Susceptible to penicillin, amoxicillin, sulbenicillin, and erythromycin.

Source: gingival crevicular fluid obtained from beagles.

DNA G+C content (*mol%*): 41–42 (T_m).

Type strain: NUM 301, ATCC 55562.

Sequence accession no. (*16S rRNA gene*): DQ677835.

Porphyromonas gulae Fournier, Mouton, Lapierre, Kato, Okuda and Menard. 2001, 1187[VP]

gu'lae. L. n. *gula* [animal] mouth; L. gen. n. *gulae* from the animal mouth, referring to its isolation from subgingival plaque of various animal hosts.

The description is from Fournier et al. (2001). *Porphyromonas gulae* encompasses the "animal *Porphyromonas gingivalis*". Cellular and colonial morphology and other characteristics are as described for the genus and as given in Table 2. Twenty strains were susceptible to amoxicillin and amoxycillin/clavulanate.

Source: subgingival plaque samples of various mammals, including cat, dog, coyote, wolf, bear, and non-human primate, such as squirrel monkey and spider monkey. *Porphyromonas gulae* is the most prominent species of the genus *Porphyromonas* to be found in the oral cavity of mammals.

DNA G+C content (*mol%*): 51 (T_m).

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Type strain: Loup 1, ATCC 51700, CCUG 47701, NCTC 13180.

Sequence accession no. (*16S rRNA gene*): AF208290.

Porphyromonas levii (Johnson and Holdeman 1983) Shah, Collins, Olsen, Paster and Dewhirst 1995, 586^{VP} (*Bacteroides levii* Johnson and Holdeman 1983, 15^{VP})

lev'i.i. N.L. gen. masc. n. *levii* of Lev, named after Meir Lev, the American-English microbiologist, who first isolated this organism.

The description is from Shah et al. (1995), Holdeman et al. (1984), and Summanen et al. (2005). Cellular and colonial morphology and other characteristics are as described for the genus and as given in Table 2. Protoheme is the major porphyrin produced, but traces of protoporphyrin also occur. Succinate stimulates growth and can replace the requirement for heme. Few sugars, such as glucose and lactose, are weakly fermented. Glucose broth cultures are turbid with smooth sediment and a final pH of 5.5. Most other commonly occurring sugars, such as arabinose, cellobiose, melezitose, melibiose, raffinose, rhamnose, ribose, salicin, sucrose, trehalose, and xylose, are not fermented. Some amino acids, such as asparagine, tryptophan, phenylalanine, and glutamine, are utilized. Proteases are present. Starch is not hydrolyzed.

Porphyromonas levii is genetically distant from *Porphyromonas somerae*; however, these two species cannot readily be differentiated by phenotypic tests. Although both species contain major $C_{15:0}$ iso, $C_{15:0}$ anteiso acids, the cluster analysis of the cellular fatty acid profiles (Euclidian distance of principal components accounting for the greatest variance of the organisms) differentiates the two species.

Source: bovine rumen, cattle horn abscess, bovine summer mastitis, and bovine necrotizing vulvovaginitis.

DNA G+C content (*mol%*): 46–48 (T_m).

Type strain: LEV, ATCC 29147, CCUG 21027, CCUG 34320, HAMBI 467, NCTC 11028, VPI 10450, VPI 3300.

Sequence accession no. (*16S rRNA gene*): L16493.

Porphyromonas macacae (Slots and Genco 1980) Love 1995, 91^{VP} [*Bacteroides melaninogenicus* subsp. *macacae* Slots and Genco 1980, 84^{VP}; *Bacteroides macacae* (Slots and Genco 1980) Coykendall, Kaczmarek and Slots 1980, 563^{VP}; *Porphyromonas salivosa* Love, Bailey, Collings and Briscoe 1992, 438^{VP}]

ma.ca'cae. N.L. fem. n. *Macaca* genus name of the macaque; N.L. gen. n. *macacae* of the macaque.

The description is from Love et al. (1995) and Holdeman et al. (1984). After 6 d on blood agar plates, colonies are 0.1–0.2 mm, entire, dome shaped, and creamy brown. After 9 d in the presence of *Staphylococcus epidermidis*, however, the colonies are 1.0–1.5 mm in diameter, entire, umbonate with a central depression. As incubation progresses, the surfaces of colonies may become wrinkled with multiple central depressions and peripheral ridging. Gelatin is hydrolyzed. Lipase is not produced. Other characteristics are as described for the genus and as given in Table 2.

Porphyromonas macacae ATCC 33141 can be distinguished by phenotypic criteria from cat strains, suggesting that cat and monkey biovars exist. (See discussion of *Porphyromonas salivosa* under *Taxonomic note*, below.)

Source: oral cavities, subcutaneous abscesses, and pyothoraxes of animals, including cats and monkeys. Important pathogen in animal bite infections in humans.

DNA G+C content (*mol%*): 43–44 (T_m).

Type strain: 7728-L6C, Slots' strain 7728-L6C, ATCC 33141, CCUG 47703, DSM 20710, NCTC 13100.

Sequence accession no. (*16S rRNA gene*): L16494.

Taxonomic note: Strain NCTC 11632^T of *Porphyromonas salivosa* (Love et al., 1987) Love et al., 1992, 438^{VP} exhibits 99.3% rRNA gene sequence homology (Paster et al., 1994) and a mean level of DNA–DNA hybridization of 81% (Love, 1995) with *Porphyromonas macacae* ATCC 33141^T; therefore, the species *Porphyromonas salivosa* was included in the species *Porphyromonas macacae* by Love (1995). However, *Porphyromonas macacae* ATCC 33141^T can be distinguished by

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phenotypic criteria from cat strains (*Porphyromonas salivosa*), suggesting that cat and monkey biovars of *Porphyromonas macacae* exist. The members of the cat biovar have different colonial morphologies on blood agar: the colonies are 0.5–1.5 mm in diameter, circular, entire, dome-shaped, and brown–black at 72 h on blood agar plates. Furthermore, unlike *Porphyromonas macacae* ATCC 33141^T, the cat biovars produce different whole-cell protein and proteinase profiles on SDS-PAGE gels, and are lipase-positive and sorbitol-negative.

Porphyromonas salivosa (Love, Johnson, Jones, and Calverley 1987) Love, Bailey, Collings and Briscoe 1992, 438^{VP} = *Porphyromonas macacae* (senior heterotypic synonym) (*Bacteroides salivosus* Love, Johnson, Jones and Calverley 1987, 308)

sal.i.vo'sa. L. fem. adj. *salivosa* resembling saliva, slimy.

The description is from Love et al. (1995), Love et al. (1992), and Love et al. (1987). After 72 h on blood agar plates, colonies are 0.5–1.5 mm, entire, dome shaped, and brown–black. Gelatin is hydrolyzed. Lipase is produced.

Porphyromonas salivosa NCTC 11632^T can be distinguished by phenotypic criteria from monkey strains (*Porphyromonas macacae*), suggesting that cat and monkey biovars exist. (See *Taxonomic note*, above, under *Porphyromonas macacae*.)

DNA G+C content (*mol%*): 42–44 (T_m).

Type strain: ATCC 49407, NCTC 11632, VPB 157, CCUG 33478.

Sequence accession no. (*16S rRNA gene*): L26103.

Porphyromonas somerae Summanen, Durmaz, Vaisanen, Liu, Molitoris, Eerola, Helander and Finegold 2006, 925^{VP} (Effective publication: Summanen, Durmaz, Vaisanen, Liu, Molitoris, Eerola, Helander and Finegold 2005, 4458.)

so'mer.ae. N.L. gen. fem. n. *somerae*, of Somer, named in honor of the late Finnish microbiologist Hannele Jousimies-Somer.

The description is from Summanen et al. (2005). Colonies incubated on blood agar for 2 d often exhibit a "patchy" growth pattern, with larger colonies surrounded by smaller colonies; they are circular, entire, and convex. The colonies on laked rabbit blood agar are white–yellow to tan; after 4 d of incubation, the colonies are pigmented (light brown to dark brown) and show no or occasionally weak red fluorescence under long-wave UV light. Weakly saccharolytic, the pH of glucose, lactose, and maltose cultures is 5.3–5.4 after 5 d of incubation. Most other commonly occurring sugars, such as arabinose, cellobiose, melezitose, melibiose, raffinose, rhamnose, ribose, salicin, sucrose, trehalose, and xylose are not fermented. β-lactamase is produced by 21% of the strains. Some strains are resistant to clindamycin. Other characteristics are as described for the genus and as given in Table 2.

Porphyromonas somerae is genetically distant from *Porphyromonas levii*; however, these two species cannot readily be differentiated by phenotypic tests. Although both species contain major $C_{15:0}$ iso, $C_{15:0}$ anteiso acids, the cluster analysis of the cellular fatty acid profiles (Euclidian distance of principal components accounting for the greatest variance of the organisms) differentiates the two species.

Source: various clinical specimens of non-oral origin, mainly from chronic foot infections of diabetics or other patients with vascular insufficiency.

DNA G+C content (*mol%*): 47.8 (T_m).

Type strain: WAL 6690, ATCC BAA-1230, CCUG 51464.

Sequence accession no. (*16S rRNA gene*): AY968205.

Porphyromonas uenonis Finegold, Vaisanen, Rautio, Eerola, Summanen, Molitoris, Song, Liu and Jousimies-Somer 2005, 547^{VP} (Effective publication: Finegold, Vaisanen, Rautio, Eerola, Summanen, Molitoris, Song, Liu and Jousimies-Somer 2004, 5301.)

ue.no'nis. N.L. gen. masc. n. *uenonis* of Ueno, in honor of the late Japanese microbiologist Kazue Ueno.

The description is from Finegold et al. (2004). Growth is stimulated by 5% horse serum and similar additives. Cellular and colonial morphology and other characteristics are as described for the genus and as given in Table 2. *Porphyromonas uenonis* can be distinguished from *Porphyromonas asaccharolytica* by phenotypic tests (Table 2); however, DNA–DNA reassociation studies are required to distinguish these two species genetically. Susceptible to most antimicrobial agents. Some strains produce β-lactamase.

Source: part of a mixed flora in various infections, which apparently have their origin in the intestinal tract. The habitat is probably the human gut.

DNA G+C content (*mol%*): 52.5 (T_m).

Type strain: WAL 9902, ATCC BAA-906, CCUG 48615.

Sequence accession no. (*16S rRNA gene*): AY570514.

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Other organisms

Porphyromonas bennonis, a novel species isolated from human clinical specimens, was recently published in IJSEM (Summanen et al., 2009).

DNA G+C content (mol%): 58 (T_m).

Type strain: WAL 1926C, ATCC BAA-1629, CCUG 55979.

Sequence accession no. (16S rRNA gene): EU414673.

The 16S rRNA gene sequence of an organism called "*Porphyromonas canis*" was published in GenBank with accession number AB0034799. "*Porphyromonas canis* sp. nov. isolated from dog". Unpublished. Strain ATCC 55562^T of *Porphyromonas gingivicanis* Hirasawa and Takada 1994, 640^{VP} exhibits 99.9% rRNA gene sequence homology with "*Porphyromonas canis*" JCM 10100, suggesting that *Porphyromonas canis* is a later synonym of *Porphyromonas gingivicanis*.

No valid description of the species "*Porphyromonas canis*" can be found in the literature. However, the following information is available.

DNA G+C content (mol%): 52.5 (T_m).

Type strain: JCM 10100.

Sequence accession no. (16S rRNA gene): AB034799.

Taxonomic note: Hardham, Dreier, Wong, Sfintescu and Evans (2005) mentioned an organism named "*Porphyromonas denticanis*". They indicated that *Porphyromonas salivosa*, "*Porphyromonas denticanis*" (a novel species), and *Porphyromonas gulae* were the most frequently isolated black-pigmented anaerobic bacteria associated with canine periodontitis. However, no valid description of "*Porphyromonas denticanis*" can be found in the literature, and there are no GenBank entries for it.

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